Eldridge Surianto

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EDUCATION

Georgia Institute of Technology **Atlanta, Georgia**

B.S. in Computer Engineering | Minor in Applications of AI/ML | GPA: 3.95 *Expected Graduation: May 2027*

- **Concentrations:** Systems and Architecture & Computing Hardware and Emerging Architectures
- **Related Coursework:** Computer Architecture, VLSI & Advanced Digital Design, Semiconductor Physics, Computer Organization, Signal Processing, Digital Design Laboratory, Data Structures & Algorithms, Web Design

SKILLS

Programming: C/C++, Python, Rust, SystemVerilog, Verilog, VHDL, RISC-V, MIPS, MATLAB, Java, HTML, CSS, Javascript
Technical: FPGAs, HLS, IC Design, Data Structures & Algorithms, Finite Element Analysis, Web Development, Machine Vision
Tools: Cadence Virtuoso, Vitis HLS, Vivado, ModelSim, Quartus Prime, iVerilog, GTKWave, Linux CLI, Git, Django, MATLAB, COMSOL
Professional Organizations: SiliconJackets, Yellow Jackets Space Program
Platforms: Linux (Ubuntu, RHEL), Windows, MacOS **Languages:** English, Chinese, German

PROFESSIONAL EXPERIENCE

Software Engineer | LightningX | Sharc Lab | Georgia Tech **August 2025 - Present**

- Refactored waveform tracing backend for a Rust-based High-Level Synthesis (HLS) simulation framework, reducing execution time by 50% with an optimized waveform output module, enabling efficient reverse engineering of complex HLS designs.
- Automated APATB generation using a Python script, leveraging Clang AST and LLVM bitcode to extract kernel-level code, enabling the creation of automatic test benches for validating HLS-generated hardware designs.

Software Intern | Automation Technology Center PTE LTD | Singapore **May 2025 - August 2025**

- Engineered a high-performance 3D Laser Scanning system using OpenCV for image processing, OpenGL for interactive 3D visualization, and multithreading/OpenMP to reduce processing time for 500 images from 60s to <2s.
- Leveraged EtherCAT (SOEM) and Yaskawa SGD7S servo drive for real-time encoder-triggered camera control via Spinnaker SDK, enabling precise, automated lead frame inspection and integration into semiconductor fabrication machines.

Undergraduate Research Assistant | Plasma & Dielectrics Lab | Georgia Tech **August 2024 - August 2025**

- Developed MATLAB scripting pipelines to reduce Finite Element Analysis (FEA) setup time in COMSOL by over 99%.
- Executed thousands of FEA simulations to analyze impact of material inhomogeneity on current distribution using the Monte Carlo method. Presented poster on research findings and methodology at Lester Eastman Conference 2025.

PROJECTS

Pipelined RISC-V Processor

- Implemented a 5-stage pipelined 32-bit RISC-V (RV32I) processor in SystemVerilog, implementing hazard detection, stall control, data forwarding, and branch flushing to minimize pipeline bubbles and maximize instruction throughput.
- Debugged and optimized the design with ModelSim, Icarus Verilog, and GTKWave, building custom testbenches to ensure functional correctness, performance efficiency, and compliance with RV32I ISA specifications.

VHDL RTL LED Metronome on DE10-Standard ModelSim-Intel SoC FPGA

- Designed a VHDL-based LED metronome peripheral on a DE10-Standard FPGA using Quartus Prime and ModelSim timing simulations, incorporating a state machine to control the PWM-controlled LED brightness and clock division for adjustable BPM.
- Integrated the peripheral into a custom SCOMP (Simple Computer) architecture, enabling dynamic tempo and LED control via SCOMP assembly instructions directly interfacing with memory-mapped registers.

Wordle Game Embedded Systems Programming on Mbed uLCD kit

- Developed a Wordle-inspired game in C++ for an ARM Cortex-M 32-bit microcontroller, achieving smooth real-time gameplay by optimizing memory usage and code efficiency on a resource-constrained system using GPIO and interrupts.
- Integrated dynamic visuals, audio, and responsive input handling by applying real-time debugging techniques and precise timing control, improving user responsiveness and strengthening hardware-software interaction across all gameplay states.

LEADERSHIP EXPERIENCE

Peer Leader | ECE Discovery Studio | Georgia Tech **August 2025 - Present**

- Mentored 15 first-year ECE students in an extended orientation course, providing guidance on academic and social opportunities on campus and organizing activities on professional development like elevator pitches and resume writing.
- Managed course logistics including attendance, discussion, and grading assignments such as resumes and course registration plans; facilitated student project showcase meetings, including a technical article analysis and a personal ECE project.

Product Owner | CS2340 Objects and Design | Georgia Tech **August 2024 - December 2024**

- Led a 5-person team, strategically translating project requirements into actionable tasks and allocating responsibilities, while overseeing UI/UX and backend implementation to create Django web apps like a Food Finder and Spotify Wrapped app.
- Applied SCRUM Agile principles to manage member progress and focus on user-centric features in an engaging format.